



USING BINOCULARS

Whatever size of binoculars astronomers choose, it can be difficult to keep them steady. Placing elbows against something solid, such as a wall, or sitting down in a lawn chair, can help stop the binoculars from wobbling. Giant binoculars are too heavy to hold steady in the hands, so should be supported on a tripod. Another common problem is finding the target object in the field

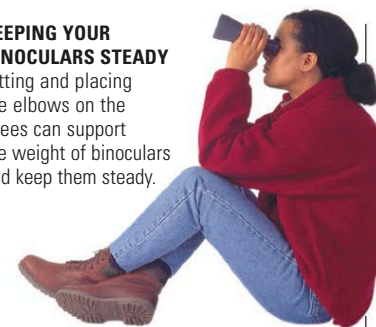
of view, even when the object is visible to the naked eye. One method is to establish the position of the target in relation to an easier-to-locate object, then locate the easier object and finally navigate to the target object.

GIANT BINOCULARS
Dedicated astronomers generally prefer binoculars with objective lenses of 70 mm (2.8 in) and magnifications of 15–20 $\times$.

Alternatively, work upward from a recognizable feature on the horizon.

KEEPING YOUR BINOCULARS STEADY

Sitting and placing the elbows on the knees can support the weight of binoculars and keep them steady.



HOW TO FOCUS A PAIR OF BINOCULARS

A pair of binoculars is not immediately in perfect focus for every user, since users' eyesight differs. To fix this, follow the instructions below.

1 IDENTIFY FOCUSING RING

Find which eyepiece can be rotated to focus independently (usually the right). Look through with your eye closed on that side.

2 FOCUS LEFT EYEPIECE

Rotate the binoculars' main, central focusing ring, which moves both eyepieces, until the left-eyepiece image comes into sharp focus.

3 CLOSE LEFT EYE, OPEN RIGHT EYE

Now open only the other eye (in this example, the right), and use the eyepiece focusing ring to bring the image into focus.

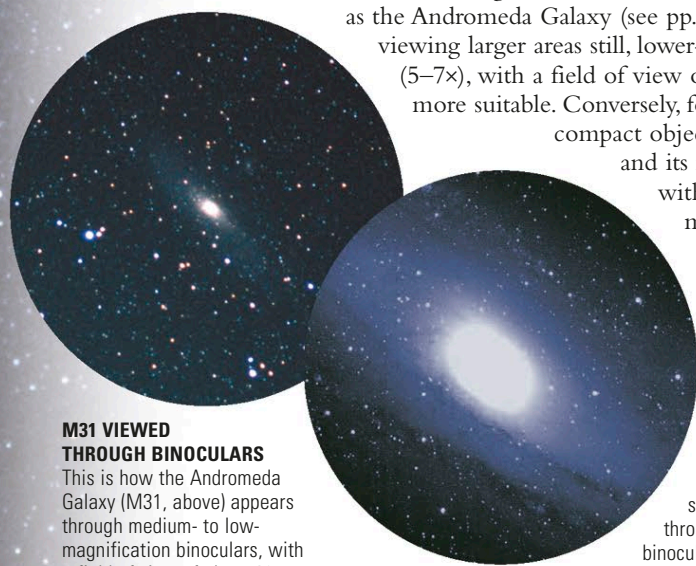
4 FOCUS AND THEN USE BOTH EYES

Both eyepieces should now be in focus, so now you can open both eyes and start observing.

BINOCULAR FIELD OF VIEW

The size of the circular area of sky seen through binoculars is called the field of view and is usually expressed as an angle. The field of view is closely related to magnification—the higher the magnification, the smaller the field of view. A typical field of view of a pair of medium-power binoculars (10 $\times$) is 6–8 $^\circ$.

This offers a good compromise between adequate magnification and a field of view wide enough to see most of a large object such as the Andromeda Galaxy (see pp.312–13). For viewing larger areas still, lower-power binoculars (5–7 $\times$), with a field of view of at least 9 $^\circ$, are more suitable. Conversely, for looking at more compact objects, such as Jupiter and its moons, binoculars with higher magnification, and a field of view of 3 $^\circ$ or even less, are better to use.



M31 VIEWED THROUGH BINOCULARS

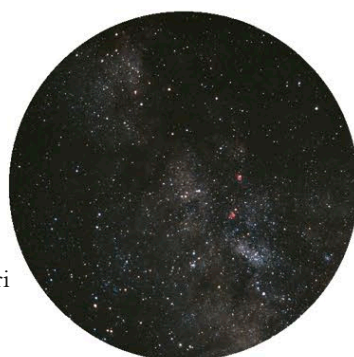
This is how the Andromeda Galaxy (M31, above) appears through medium- to low-magnification binoculars, with a field of view of about 8 $^\circ$.

M31 VIEWED THROUGH A TELESCOPE

Here the central part of the Andromeda Galaxy is shown as you might see it through very-high-magnification binoculars, or a small telescope, with a field of view of about 1.5 $^\circ$.

BINOCULAR OBJECTS

A striking first object for a novice binocular user is the Orion Nebula (see p.241). Other choices might be the Andromeda Galaxy (above), and the fabulous star clouds and nebulae in the Sagittarius and Scorpius regions of the Milky Way, including the Lagoon Nebula (see p.243). For viewers south of 50 $^\circ$ N, an excellent binocular object is the Omega Centauri star cluster (see p.294). To find these, all that is needed is some star charts (see pp.426–501) or an astronomy app on your phone. Also try observing the Moon, Jupiter and its moons, and the phases of Venus.



THE MILKY WAY

Shown here is a dense region of the Milky Way in Sagittarius, as seen through low-power binoculars with a field of view of 12 $^\circ$.



ORION NEBULA

This appears as a blue-green smudge in Orion, shown here as it appears in medium-power binoculars with a field of view of 8 $^\circ$.



THE PLEIADES

This spectacular star cluster in Taurus is seen here as it appears through high-power binoculars with a field of view of about 3 $^\circ$.